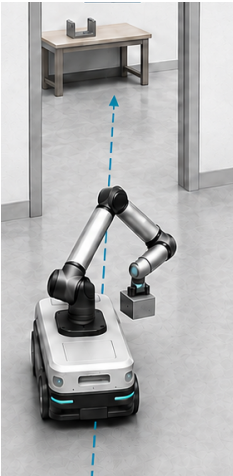
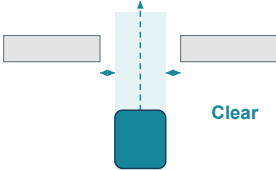
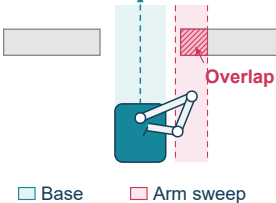
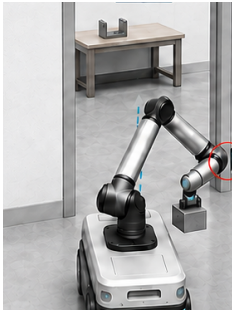
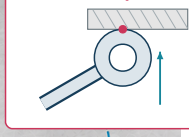


Asimov Cascade: local correctness does not guarantee safe composition

TYPE	C1 Local correctness	C2 Interface mismatch	C3 Downstream propagation	C4 Unsafe trajectory
MEANING	Modules meet local contracts under their own assumptions: base planning, pose holding and path tracking.	A guarantee covering only the base footprint is consumed as a guarantee for the whole robot.	The scope error passes through task execution into trajectory generation and base control.	The composed motion is unsafe: the base clears the opening, but the held elbow strikes the jamb.
ILLUSTRATIVE EXAMPLE	Deliver a workpiece 	Base-only check  Whole-robot check  □ Base □ Arm sweep	Base planner Base footprint only ↓ Task executive Whole-robot assumption ↓ Trajectory generator Timed path ↓ Base controller Track motion Error crosses downstream interfaces	Elbow-jamb collision  Contact · top view 

Hypothetical example · four defining conditions, not four time steps